

Analyzing the Impact of Economic Factors on Housing Prices in the United States

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Abstract— Economic conditions vary across U.S. states, and those differences can shape housing prices, migration patterns, and overall affordability, which are important for policymakers and researchers. This study examines the relationship between housing prices and multiple economic factors, including income growth, population change, migration, unemployment rates, and cost-of-living differences, across all 50 states in the United States from 2020 to 2024. To do this, data from the Bureau of Labor Statistics, the U.S. Census Bureau, the Bureau of Economic Analysis, the Federal Housing Finance Agency, and the Internal Revenue Service were merged into a unified state-year dataset. After the data was cleaned and organized, exploratory data analysis was used to identify patterns, correlations, and trends across states and over time. A two-way fixed-effects panel regression was then used as the main model to examine how these variables are associated with housing prices, while controlling for state-level and year-level effects. Random Forest Regression was used as a comparison model. The results suggest that Regional Price Parities (RPP) remained one of the strongest positive factors associated with housing prices, while unemployment remained negatively associated with housing prices. Income change and population contributed to the broader housing-price story, while migration appeared more limited because of shorter data coverage. These findings suggest that housing prices are shaped by multiple economic factors rather than any single variable.

Index Terms— housing prices, migration, unemployment, income growth, cost of living, Regional Price Parities, panel regression, fixed effects, state-level economic analysis.

I. INTRODUCTION

Housing prices matter because they affect more than just the cost of buying a home. They also shape affordability, migration decisions, financial stability, and the overall economic pressure people feel in different parts of America. When housing prices rise, it becomes harder for many people to afford to live in certain areas, and that can influence where they

choose to move, work, or stay. As a result, studying housing prices is important for understanding not just the housing market itself but also broader economic conditions across the United States.

However, economic conditions vary across states. Some states experience stronger income growth, lower unemployment, or larger population increases, while others face the opposite, such as slower growth, higher unemployment, or higher overall living costs. These differences can affect housing demand in different ways, so it's imperative to have that understanding when collecting data. A state with stronger job growth and rising incomes may see more pressure on housing prices, while a state with weaker economic conditions may not experience the same kind of price increases. Looking at these differences across states makes it easier to see why housing prices do not move the same way everywhere. Understanding these differences is important when accounting for potential bias in the analysis.

This study focuses on several variables that may help explain those differences: unemployment rates, income growth, population change, migration patterns, and cost-of-living differences. Housing prices are measured using the Federal Housing Finance Agency House Price Index (HPI), which is an index of relative housing-price movement over time rather than a dollar measure of home prices. Regional Price Parities (RPP) are used to represent cost-of-living differences across states. RPP is also an index, where values around 100 represent the national average price level, values above 100 indicate a higher-than-average cost of living, and values below 100 indicate a lower-than-average cost of living. Therefore, neither HPI nor RPP should be interpreted as dollar amounts. Together, these variables provide a broader picture of the economic and demographic factors that may be associated with housing price patterns across states.

The core question of this research is: **How are unemployment, income growth, population change, migration, and cost-of-living differences associated with housing prices in U.S. states between 2020 and 2024?** This study also seeks to

determine which of these variables exhibits the strongest relationship with housing prices, reinforcing the idea that a combined, rather than isolated, approach best explains variation in housing prices.

To answer these questions, this project combines data from the Bureau of Labor Statistics, the U.S. Census Bureau, the Bureau of Economic Analysis, the Federal Housing Finance Agency, and the Internal Revenue Service into a unified state-year dataset. Because the study period begins in 2020, the analysis also takes place during a period of unusual housing market conditions in the United States. After cleaning and merging the data, exploratory data analysis is used to identify trends and relationships across states and over time. The study then applies a two-way fixed effects panel regression as the main model to examine how these economic factors jointly relate to housing prices while accounting for repeated state observations over time. Random Forest Regression is used as a comparison model. This approach supports the project's main argument that housing prices are better understood through a multivariable framework rather than a one-variable explanation.

II. LITERATURE REVIEW

Prior research shows that housing prices are influenced by a mix of economic and demographic factors rather than one single variable. Glaeser, Gyourko, and Saks argue that housing prices rise because of broader housing market pressures and structural constraints, while Duca, Muellbauer, and Murphy show that house price cycles reflect wider economic, demographic, and market conditions [5], [11]. These studies are important for this project because they support the idea that housing prices should be studied through a multivariable framework instead of being explained by only one factor at a time.

Other research suggests that migration, labor market conditions, and affordability are also important parts of the housing story. Prior work shows that higher income can increase purchasing power and housing demand, while weaker labor conditions can reduce income stability and lower housing demand. Molloy, Smith, and Wozniak show that internal migration in the United States is closely tied to broader economic conditions, which helps explain why migration belongs in this project even if its role may differ across states [8]. Cost-of-living differences also matter because states with higher overall price levels often face higher housing costs, which is why

Regional Price Parities are useful in this study as a measure of cost of living.

Another important factor in this project is the COVID-19 period itself, since the dataset begins in 2020. Liu and Su found that the pandemic changed housing demand in the United States by reducing demand for density, which suggests that housing preferences shifted during this time [21]. Mondragon and Wieland also argue that remote work played a major role in the rise of housing prices, showing that the shift to remote work explains over half of the increase in U.S. real house prices from 2019 to 2023 [22]. These studies matter for this project because they suggest that part of the housing price growth observed from 2020 onward may reflect pandemic-related changes in where people wanted to live, how much housing space they wanted, and how remote work changed housing demand across regions.

Taken together, the literature suggests that housing prices are influenced by income, labor market conditions, migration, and broader cost-of-living differences. However, these factors are often discussed separately. This project adds to that literature by combining unemployment, income growth, population, migration, housing prices, and Regional Price Parities into one 50-state panel dataset covering 2020 to 2024. By doing that, the study examines how these variables relate to housing prices when they are analyzed together during a period that also includes the unusual housing conditions associated with COVID-19.

III. METHODOLOGY

This study uses a quantitative state-year panel design. The final sample includes only the 50 U.S. states and removes data on the District of Columbia from the merged analysis. The core panel covers 2020 through 2024 and includes the following variables: FHFA house price index, BLS unemployment rate, BEA-based income change, Census population, and BEA Regional Price Parities. Migration is added using IRS state-to-state inflow and outflow data for the 2020–2021, 2021–2022, and 2022–2023 flow years. In the merged panel, these files are used to create migration observations for 2021, 2022, and 2023. As a result, the full-panel main model uses unemployment rate, income change, population, and RPP, while a reduced 2021 to 2023 specification adds net migration.

The first step in the method is data cleaning. Each source file was standardized so the records could be merged by state and year. State names were aligned

across files, the United States total row was removed where needed, and the District of Columbia was removed to keep the sample at 50 states consistent. The RPP file was reshaped from wide format to long format so that each row holds one state, one year, and one RPP value. The migration files were aggregated to the state level and used to create inflow, outflow, and net migration measures.

The merged dataset is organized as a panel. Because the same 50 states were observed repeatedly across multiple years, panel regression was appropriate for the analysis. The two-way fixed effects approach allowed the study to control for both state-level characteristics and year-level economic shocks while examining the relationship between economic variables and housing prices.

For the full 2020 to 2024 panel, there are 250 state-year observations. For models or charts that use migration, the usable sample is smaller because IRS migration data are only available for the 2020–2021 through 2022–2023 flow years. In the merged panel, that means migration can only be analyzed for 2021 through 2023. That leaves 150 observations for the migration subset. This matters because any statement about migration must be limited to the years observed in the IRS files.

Exploratory data analysis was used first. Time-series plots were used to examine average changes over time; scatterplots were used to compare housing prices with RPP and migration; and a correlation heatmap was used to show the strength and direction of pairwise relationships. The cleaned 50-state results show that higher-RPP states tend to have higher housing price index values, suggesting that the cost of living and housing prices are closely linked, which is another reason RPP is included in the baseline regression rather than treated solely as a descriptive variable. The results also show that higher unemployment tends to line up with lower housing price index values. Income change has a positive relationship with housing prices, but not as strong as RPP. Population by itself is much weaker in the simple correlation results. Migration was included as a factor for the available years, but its relationship with housing prices is weaker than that of RPP. To move beyond descriptive analysis, the project applies a two-way fixed effects panel regression as the main model. Because the same 50 states are observed repeatedly from 2020 to 2024, panel regression is more appropriate than a simple pooled regression. The model includes state fixed effects to account for stable state-level differences and year fixed effects to account for common shocks across years, including

pandemic-era effects. This approach helps reduce omitted-variable bias by controlling for unobserved factors that are not directly measured but may still affect housing prices across states and years.

The main panel specification can be written as:

$$\text{Housing Price Index} = \beta_0 + \beta_1(\text{Unemployment Rate}) + \beta_2(\text{Income Change}) + \beta_3(\text{Population}) + \beta_4(\text{RPP}) + \text{State Fixed Effects} + \text{Year Fixed Effects} + \varepsilon.$$

This specification estimates how changes in unemployment, income, population, and cost of living are associated with housing prices within states over time, rather than only comparing states to one another at a single point.

Migration remains part of the broader project, but because IRS migration data are only available for 2021 through 2023 in the merged panel, migration is treated as a supporting and more limited variable rather than the core variable in the full 2020–2024 panel model.

In addition to the panel regression, Random Forest Regression is used as a comparison model to test whether a more flexible machine learning method captures nonlinear relationships more effectively.

Comparing these models helps determine whether a more flexible machine learning method improves predictive performance beyond the traditional regression approach. There are also some important limits to this method. The project overview notes that some datasets may have missing values, inconsistent formats, or differences in reporting periods. It also highlights the challenge of distinguishing correlations from causation. That is an important point in this project because even if two variables move together, that does not automatically mean one is causing the other. Another limit is that migration does not cover the full 2020 to 2024 period, so any model that includes net migration must be limited to the reduced 2020 to 2023 subset and interpreted more carefully than the full-panel baseline model. Because of that, the analysis is meant to identify patterns and relationships, not to make overly strong causal claims. To better account for differences across states and years, this study applied a two-way fixed effects panel regression model. State fixed effects control for characteristics that remain relatively stable within states over time, such as geography, zoning environments, and long-term economic conditions. Year fixed effects control for broader national shocks that affect all states simultaneously, including

pandemic-related disruptions during the COVID-19 period.

This study has several limitations that should be kept in mind when interpreting the results. First, migration data is only available for the 2020–2021, 2021–2022, and 2022–2023 flow years, which means migration cannot be tested across the full 2020 to 2024 panel and is limited to the 2021 to 2023 portion of the merged dataset. Second, the project uses state-level data, so differences within states are not captured. Housing market patterns may look different at the city, county, or metro level. Third, the models in this study identify statistical relationships rather than causal effects. Finally, because the sample period begins in 2020, some of the observed housing price patterns may reflect unusual pandemic-era conditions rather than only longer-run housing market behavior. These limits do not remove the value of the results, but they do mean the findings should be interpreted with appropriate caution.

Overall, the methodology is designed to support a multivariable analysis of housing prices and economic conditions across U.S. states. By combining multiple datasets into a single panel structure, conducting exploratory analysis, and estimating a regression model, the project creates a framework for assessing how unemployment, income, population, migration, and cost-of-living conditions relate to housing market outcomes over time and across places.

IV. RESULTS AND ANALYSIS

Using the merged 50-state dataset, this study first examined correlations and visual patterns, then applied a two-way fixed effects panel regression as the main model. Random Forest Regression was used as a comparison model to evaluate whether a machine learning approach could improve predictive performance.

The correlation and regression results tell a consistent story. Regional Price Parities (RPP) remain the strongest associated factor in the dataset, with a correlation coefficient of about 0.56 to 0.60 with housing prices. This indicates that states with higher cost-of-living levels generally also have higher housing prices. Income change shows a positive but weaker relationship, while unemployment is negatively associated with housing prices. Population and migration show weaker effects, suggesting that they are not strong standalone explanatory variables in this dataset. These results reflect statistical

associations and do not imply causation between the variables.

The panel regression results support the same broad pattern seen in the correlations and visualizations. Regional Price Parities (RPP) remained one of the strongest positive factors associated with housing prices, suggesting that cost-of-living differences are closely tied to housing-price patterns. Unemployment remained negatively associated with housing prices, while population appeared weaker as a standalone factor. Income change contributed to the broader housing-price story, although its effect was less consistent than that of RPP.

To further compare economic relationships across time periods, regression grids were created separately for the COVID period (2020–2022) and post-COVID period (2023–2024). The visualizations show that Regional Price Parities (RPP) consistently maintained the strongest positive relationship with housing prices across both periods. Income change remained moderately positive, while unemployment generally showed weaker negative relationships. Population effects appeared smaller and less consistent across states.

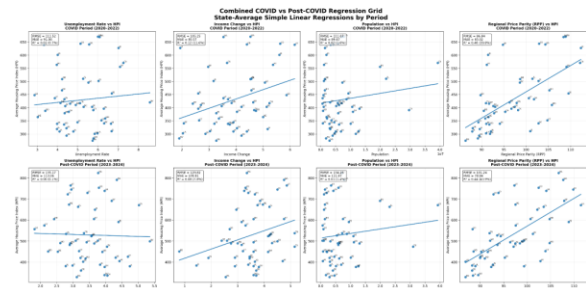


Fig. 1 Combined COVID vs Post-COVID regression grid comparing unemployment, income change, population, and Regional Price Parities against housing prices across U.S. states.

The reduced migration model for 2021 to 2023 provides a more limited test because the number of observations is smaller, and migration data is only available for that subset. In that model, net migration does not appear to be as strong as RPP, income change, or unemployment. This suggests that migration may matter more indirectly, or that its effect is harder to isolate at the state-year level in the current dataset. Taken together, the regression models support the broader conclusion that housing prices are best explained by multiple economic conditions rather than a single variable.

To compare the traditional regression approach with a machine learning method, Random Forest Regression was applied to the same data used in the baseline regression model. On the full 2020–2024 dataset using unemployment rate, income change, population, and RPP, Random Forest was used as a comparison model alongside the two-way fixed effects panel regression. The comparison suggests that model performance depends on both the size and structure of the data. Random Forest remained useful for highlighting flexible nonlinear patterns, while the panel regression was more appropriate for the repeated state-year structure of the dataset.

The Random Forest visualizations also suggest that some relationships may be nonlinear across states and time periods. Compared with the simple regression plots, Random Forest captured more flexible patterns between economic variables and housing prices, particularly for RPP and income change.

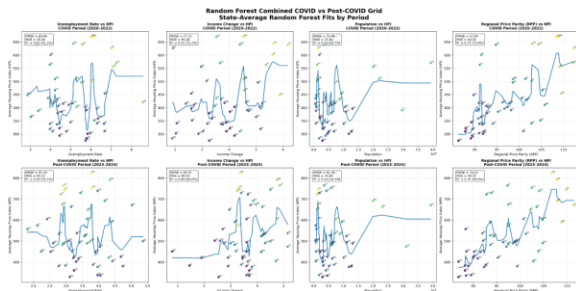


Fig. 2 Random Forest regression grid comparing economic variables and housing prices during the COVID and post-COVID periods.

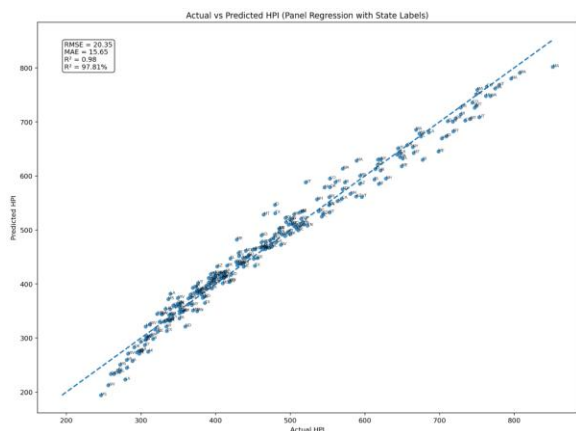


Fig. 3 compares actual and predicted housing price index values from the panel regression model. The close alignment of many points to the diagonal line suggests that the model captured a substantial share of housing-price variation across the state-year panel.

The unemployment rate is negatively associated with housing prices. Higher unemployment rates correspond to lower prices, as shown in Fig. 2. This implies that labor market downturns decrease housing demand and, hence, prices.

Migration appears to be the weakest predictor of housing prices among the variables considered in our study. Despite the general perception of population movement as a leading cause of the rise in housing demand, our findings indicate a very limited effect over the period considered. Our dataset includes IRS migration flow data for 2020–2021, 2021–2022, and 2022–2023, which are used in the merged panel for 2021 to 2023. This still makes migration more limited than the full 2020 to 2024 panel and makes it harder to analyze longer-run migration trends.

Furthermore, the time-series analysis indicates that housing prices have been rising at an average pace from 2020 to 2024. Referring to Fig. 4, the average Housing Price Index increased from about 372 to 544. This reflects a strong upward trend in housing prices across states during the study period.

Overall, the results suggest that housing prices are best explained by a combination of economic factors rather than any single variable. Cost of living, measured through RPP, stands out as the strongest associated factor, while unemployment and income provide additional explanatory power. The comparison between the two-way fixed effects panel regression and Random Forest Regression further suggests that model performance depends on both the size and structure of the data. Because RPP is an index rather than a dollar measure, higher RPP values indicate higher overall state price levels relative to the national average, not a direct dollar cost of living.

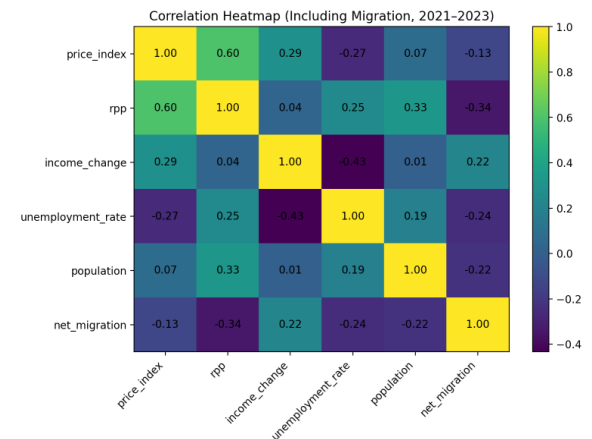


Fig. 4. Correlation heatmap of housing price index and key economic variables, including RPP, income change, unemployment rate, population, and net migration (2021–2023). The results show that RPP has the strongest positive relationship with housing prices, while the unemployment rate exhibits a negative correlation.

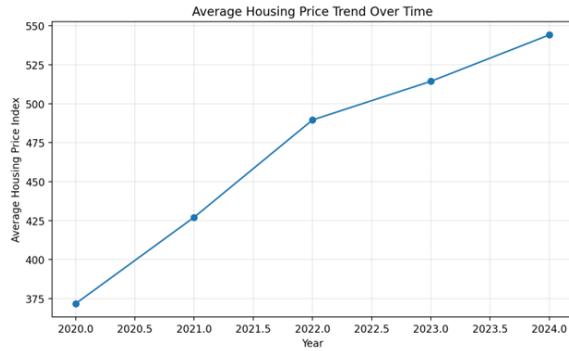


Fig. 5. Trend of average housing price index from 2020 to 2024. The steady increase reflects strong growth in housing prices across U.S. states during this period.

To further illustrate the geographic distribution of housing prices, Fig. 6 presents a choropleth map of the housing price index across U.S. states.

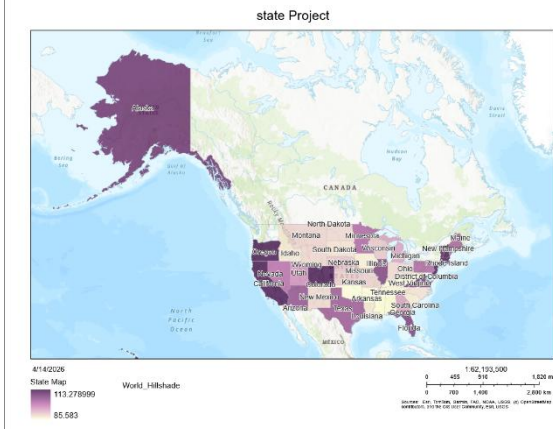


Fig. 6. Choropleth map of housing price index across U.S. states. Darker shades indicate higher housing price levels, highlighting regional disparities, with higher concentrations in coastal and economically strong states.

Overall, the panel regression and machine learning comparisons indicate that housing prices are influenced by multiple interacting economic conditions, with cost-of-living differences consistently emerging as the strongest explanatory

factor across both COVID and post-COVID periods. The research question is answered by showing that housing prices from 2020 to 2024 were associated most clearly with cost-of-living differences, while unemployment and income provided additional explanatory power. The population was weaker, and migration was more limited because of data availability. These results suggest that housing prices are best explained through a combination of economic conditions rather than a single factor.

V. DISCUSSION

The results support the main idea from the literature review that housing prices are shaped by multiple factors rather than one single variable [5], [11]. In this project, the cost of living, measured through Regional Price Parities, appears to be the clearest signal. That suggests housing prices are not just moving because of local demand in a narrow sense. They seem to be tied to a broader state-level economic environment where places with higher overall prices also tend to have higher housing prices. This helps explain why RPP remained stronger than population and migration in the data. Because the main model is a two-way fixed effects panel regression, the results should be interpreted as associations within states over time, after controlling for stable differences across states and common year-level shocks. This strengthens the analysis by making the model more appropriate for the panel structure of the data.

The results also partly support the expectation that income and labor market conditions matter for housing prices. Income change shows a positive relationship with housing prices, while unemployment shows a negative one. This fits the prediction from prior research that stronger economic conditions should be associated with stronger housing markets. At the same time, the results suggest that income growth alone does not fully explain housing price differences, since RPP still appears more important in the overall pattern.

One interesting point is that migration was weaker than expected. Based on the literature, migration should matter because population movement can affect housing demand [8]. However, in this dataset, it was not one of the strongest standalone factors. One way to interpret that is that migration may be responding to broader economic conditions that are already being captured more directly by variables like RPP, income, and unemployment, rather than acting as the main driver by itself. It is also important to remember that migration is based on IRS flow-year

files covering 2020–2021 through 2022–2023, which in the merged panel means it is only measured for the reduced 2021 to 2023 subset. Because of that, its role is more limited in this study than the other variables.

The time period of this project also matters. Because the sample begins in 2020, the results should be interpreted in the context of the COVID-19 period. Prior research suggests that the pandemic changed housing demand by reducing demand for density and increasing the importance of remote work [21], [22]. That makes this relevant to the project because part of the rise in housing prices during the sample period may reflect pandemic-related shifts in where people wanted to live, how much space they wanted, and how remote work changed housing demand across regions. This may also help explain why housing prices rose strongly over time, even when some of the other variables were weaker on their own.

Another implication of these results is that housing affordability cannot be understood by looking at housing prices alone. If the cost of living is more strongly tied to housing prices than population or migration in this dataset, then affordability pressure may be part of a broader regional economic environment rather than just a simple housing shortage story. This helps explain why some states can appear economically strong while still becoming harder to afford. In that sense, the project’s findings line up with the literature by suggesting that housing prices reflect combined market pressures, labor conditions, and price-level differences rather than one single force [5], [11].

Overall, the findings suggest that housing prices in this project are best understood through a multivariable lens. The results line up with the literature review by showing that housing prices reflect the combined effect of cost of living, labor market conditions, and income rather than any one variable by itself. In this dataset, RPP stands out the most, while income and unemployment provide additional support for the broader explanation. That means the project’s original expectation was mostly supported, but it also shows that some variables, especially migration, may be more limited or indirect than they first appear.

VI. CONCLUSION

This research examined how major economic indicators relate to housing prices across U.S. states using a 50-state panel from 2020 to 2024. The results suggest that housing prices are best understood through a multivariable framework rather than a one-

variable explanation. Among the variables in this study, Regional Price Parities (RPP) emerged as one of the strongest and most consistent factors, suggesting that cost-of-living differences are a central part of the broader housing story. Unemployment also remained negatively associated with housing prices, while income and population contributed to the broader explanation. Migration appeared more limited in the current dataset because of shorter data coverage and time-period constraints.

The comparison between the two-way fixed effects panel regression and Random Forest Regression showed that different models capture different parts of the housing story. Random Forest was useful as a comparison model for predictive patterns, while the panel regression was more appropriate for the repeated state-year structure of the data. Overall, the study suggests that housing markets reflect several connected economic conditions at once, especially the cost of living, labor market strength, and income patterns.

Future research can improve upon this study by incorporating additional factors such as housing stock, interest rates, mortgage availability, and fuller migration coverage beyond 2023. Gathering data at the city or county level could provide more detail and strengthen the validity of the findings. Future work could also extend the panel framework with additional robustness checks, alternative fixed-effects specifications, and other predictive models such as gradient boosting.

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